Undergraduate Project Report

Case Study of a Rotating-Squares Auxetic Stent

Mechanics of Solids

Student: _____

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Institution: _____

Submission date: _____

This report presents an analytical case study and a proposed numerical workflow. No new physical test or finite-element result is claimed in this document.

Abstract

This case study develops an analytical design assessment of a rotating-squares auxetic stent intended to expand radially and longitudinally through hinge rotation. The report uses the supplied geometry and a one-degree-of-freedom kinematic model to predict lattice strain, ideal Poisson's ratio, solid fraction, hinge surface strain and effective modulus. The supplied STEP file was imported and measured: it contains a 90-degree tubular sector with the inner and outer radii, while its axial extent is shorter than the design length. Literature evidence is summarized separately from predictions, including reported expansion and failure observations for polyurethane auxetic stents. No physical or finite-element results are claimed. A Python script is provided for generating pattern frames, analytical plots and a ring-opening animation. An Ansys workflow and a FreeCAD–CalculiX cross-check are specified, but their result fields remain blank pending execution. Material constants are placeholders; hinge strain, fatigue, manufacturing tolerances and clinical performance require verification.

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1 Introduction and clinical problem

An oesophageal stent is intended to maintain an open passage through a narrowed region. Ali, Busfield and Rehman describe dysphagia as a major symptom in advanced oesophageal disease and discuss the use of stents as a palliative method to restore passage and help maintain the lumen [1, pp. 527–529]. The present project is a mechanical design case study, not a clinical evaluation.

The first paper reports early complications of conventional oesophageal stenting of up to migration 6%, perforation 7%, and haemorrhage 6%. It also lists longer-term rates of bleeding 19%, tumour overgrowth 36%, migration 33%, fistula formation 5%, and bolus impaction 23% [1, p. 528].

These percentages are literature figures quoted by Ali et al.; they are not complication rates measured for the rotating-squares design in this report. The authors discuss several proposed mechanical responses: a mesh structure and synclastic deformation may help address migration, soft polymeric edges may reduce edge-related injury, and an internal graft may help cover fistulae and limit tumour ingrowth [1, p. 528]. These are design arguments, not proof of clinical benefit.

1.1 Scope of this case study

The report evaluates the geometry supplied for a rotating-squares tube, establishes ideal kinematic predictions, and defines how a later Ansys analysis could test departures from the ideal mechanism. It does not establish biocompatibility, deployment safety, patency, migration resistance, fatigue life, or clinical efficacy. Those questions require material characterization, physical testing and appropriate clinical evidence [VERIFY].

2 Why a negative Poisson's ratio may help

A conventional positive-Poisson-ratio solid becomes narrower in the transverse direction when it is stretched. An auxetic structure has a negative Poisson's ratio and becomes wider in the transverse direction under the same type of extension. Ali et al. describe the rotating-squares structure as a geometry in which linked squares rotate at their connecting hinges, producing coupled extension in two in-plane directions [1, pp. 528–529].

For a tubular stent, the useful design idea is that axial and radial dimensions can change together. Ali et al. argue that radial and longitudinal expansion could reduce tumour overgrowth and that a compliant mesh with suitable edges and an internal graft could address other failure modes [1, p. 528]. Bukhari et al. similarly motivate biaxial deployment, reporting that longitudinal actuation of an auxetic stent can accompany lateral expansion [2, pp. e196–e197]. These papers present engineering reasoning and laboratory-scale evidence; neither establishes a clinical outcome for the present geometry.

The ideal rotating-squares mechanism used below has $\nu = -1$. This is a kinematic property of the idealized mechanism, not the intrinsic Poisson's ratio of the TPU polymer. The supplied polymer value $\nu \approx 0.45$ is a provisional material input [ASSUMPTION]; the lattice response and the base-material response must therefore be kept separate.

3 Stent design and geometry

3.1 Supplied design inputs

The proposed device is a thin-walled tube made from a rotating-squares pattern. The following values are the user's design inputs; the material properties are placeholders rather than measured properties of a named TPU grade.

3.2 STEP-file inspection

The file `stent2(step)(1).STEP` was successfully imported as a solid. Its measured Cartesian bounding box is approximately $5.25 \times 5.25 \times 17.00$ mm. The mesh vertices lie at radii 4.75 and 5.25 mm from the tube axis, and their angular range is 0° to 90°. The STEP geometry is therefore a quarter-tube sector, not a complete 360° tube. Its axial extent is 17.00 mm, which is 0.30 mm shorter than the nominal approximate length of 17.3 mm.

The radial dimensions agree with the stated inner and outer radii, and the radial wall difference is $5.25 - 4.75 = 0.50$ mm. The individual square side, chamfer and ligament width were not independently dimensioned from the STEP mesh; their values remain the supplied design inputs [VERIFY].

Table 1: Supplied design inputs and derived geometric checks.

Parameter	Value	Definition / source status
Initial square angle, θ_0	30°	User-supplied design input [ASSUMPTION].
Mid-surface radius, R_m	5.00 mm	User-supplied design input [ASSUMPTION].
Wall thickness, t	0.50 mm	User-supplied; STEP radial difference confirms this [STEP FILE].
Outer / inner radius	5.25/4.75 mm	User-supplied; confirmed by the imported STEP vertex radii [STEP FILE].
Circumferential / axial cell count	$n_c = 4,\ n_a = 2$	User-supplied counts [ASSUMPTION].
Nominal length	about 17.3 mm	User-supplied approximate value [ASSUMPTION].
Square side, a	4.534 mm	User-supplied [ASSUMPTION].
Chamfer, d	0.680 mm	User-supplied [ASSUMPTION].
Ligament width, h	0.567 mm	User-supplied [ASSUMPTION].
TPU elastic modulus, E	about 26 MPa	Placeholder; not a certified grade value [ASSUMPTION, VERIFY].
TPU material Poisson ratio, ν_m	about 0.45	Placeholder [ASSUMPTION, VERIFY].

Figure placeholder

Add file: `stent_cad_view.png`

Export an isometric CAD view of the imported STEP sector or the assembled full tube.

Figure 1: CAD figure slot: imported STEP geometry; the supplied file contains a quarter-tube sector [STEP FILE].

3.3 How the flat pattern wraps into a tube

The flat pattern is wrapped so that its circumferential direction becomes the tube circumference and its axial direction becomes the stent length. Using the supplied mid-surface radius, the target mid-surface circumference is $C = 2\pi R_m = 31.416\,\text{mm}$. If the four circumferential cells are repeated uniformly, the nominal circumference allocated to each repeat is $C/n_c = 7.854\,\text{mm}$.

The wrapping operation must preserve the intended hinge locations and ligament widths. A small seam or periodic interface may be present depending on the manufacturing route [VERIFY]. The sketch below illustrates the wrapping concept only; it is not a manufacturing drawing and is not to scale.

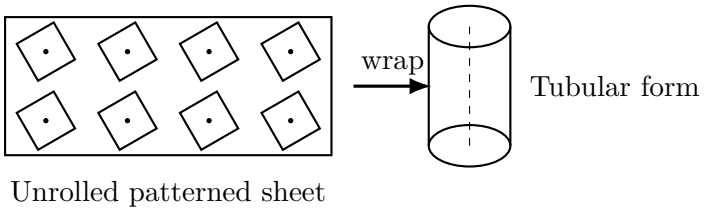


Figure 2: Schematic of the flat-to-tube wrapping concept (illustrative, not to scale). The actual STEP file supplied for this report is a quarter-tube sector, not the full tube.

4 Analytical model

4.1 Kinematic relation and ideal Poisson's ratio

Let a be the side length of a square and θ its rotation angle. Under the supplied model, the centre-to-centre separation of neighbouring squares is $D(1,1)$, where

$$D = a\cos\theta. \tag{1}$$

The geometry is governed by one rotational degree of freedom: changing θ changes the lattice pitch through $\cos\theta$, while the square side remains fixed. This rotating-squares mechanism follows the ideal kinematic model introduced by Grima and Evans [4]. At the reference angle θ_0, the pitch component is $D_0 = a\cos\theta_0$. Therefore, the nominal strain in either lattice direction is

$$\varepsilon_x = \varepsilon_y = \varepsilon = \frac{D - D_0}{D_0} = \frac{\cos\theta}{\cos\theta_0} - 1. \tag{2}$$

The two in-plane strains are equal. The ideal kinematic Poisson ratio is consequently

$$\nu_{\text{ideal}} = -\frac{\varepsilon_y}{\varepsilon_x} = -1. \tag{3}$$

This ideal result assumes rigid square units, flexible hinges, no ligament stretching, no out-of-plane distortion and no geometric interference [ASSUMPTION]. The finite-size paper by Gatt et al. warns that periodic-cell predictions need care when applied to a small number of repeat units [5].

4.2 Solid fraction and maximum opening

The supplied solid-fraction expression is

$$\phi_s(\theta) = \frac{1}{2\cos^2\theta}. \tag{4}$$

At $\theta = 45°$, $\phi_s = 1$, corresponding to the stated fully closed limit. If opening proceeds from the reference angle θ_0 to 0°, the maximum nominal opening strain is

$$\varepsilon_{\max} = \frac{1}{\cos\theta_0} - 1. \tag{5}$$

Table 2: Analytical predictions for different reference angles. The solid fraction is evaluated at the reference angle.

Reference angle θ_0	$\varepsilon_{\max}$	$\phi_s(\theta_0)$	D_0 for $a = 4.534\,\text{mm}$
20°	6.42%	0.566	4.26 mm
30°	15.47%	0.667	3.927 mm
40°	30.54%	0.852	3.473 mm
45°	41.42%	1.000	3.206 mm

For the actual reference angle of 30°, the ideal model predicts a maximum opening strain of 15.47% on moving to 0°. The model's diameter ratio for opening from a closed angle θ_{closed} to θ_{open} is

$$\frac{D_{\text{open}}}{D_{\text{closed}}} = \frac{\cos\theta_{\text{open}}}{\cos\theta_{\text{closed}}}. \tag{6}$$

From 45° to 0°, this ratio is $\sqrt{2} = 1.4142$, so a 10 mm crimped diameter would become about 14.14 mm. This example does not use the present design's 30° reference state. For the supplied 30° reference angle, opening to 0° gives a ratio of $1/\cos 30° = 1.1547$; a 10 mm diameter at 30° would therefore become about 11.55 mm. The two ratios describe different angular intervals and must not be interchanged.

4.3 Ligament bending strain and design rule

Let L_h be the free ligament length and h its width. Under the supplied pure-bending approximation, the ligament surface strain is

$$\varepsilon_{\text{lig}} = \frac{|\theta - \theta_0|h}{L_h}, \tag{7}$$

where the angle difference is in radians. Rearranging the allowable-strain condition gives the design rule

$$\frac{L_h}{h} \geq \frac{|\Delta\theta|}{\varepsilon_{\text{allow}}}. \tag{8}$$

For the supplied design,

$$D_0 = a\cos 30^\circ = 3.927\,\text{mm}, \tag{9}$$

$$L_h = \sqrt{2}\,d\cos 30^\circ = 0.833\,\text{mm}, \tag{10}$$

$$\frac{L_h}{h} = \frac{0.833}{0.567} = 1.47. \tag{11}$$

For opening from 30° to 0°, $|\Delta\theta| = \pi/6$ radians and the predicted ligament surface strain is

$$\varepsilon_{\text{lig}} = \frac{(\pi/6)(0.567)}{0.8328} = 0.3565 \approx 35.65\%. \tag{12}$$

For motion from 30° to the fully closed limit of 45°, the same model gives approximately 17.82% ligament surface strain.

The ratio $L_h/h = 1.47$ is low relative to the angular rotation required by the ideal opening motion. The model therefore predicts substantial surface strain at the hinge. This is a warning from the simplified bending calculation, not a material failure prediction. No allowable TPU strain, stress–strain curve, fatigue limit or process-dependent hinge-strength value has been supplied. Consequently, the design cannot be passed or failed against a strain limit until $\varepsilon_{\text{allow}}$ is established from a specified TPU grade and test data [VERIFY].

4.4 Effective modulus near the reference angle

The supplied near-reference effective modulus is

$$E_{\text{eff}} = \frac{Eh^3}{3L_h a^2 \sin^2\theta_0}. \tag{13}$$

Substituting the placeholder $E = 26\,\text{MPa}$, $h = 0.567\,\text{mm}$, $L_h = 0.8328\,\text{mm}$, $a = 4.534\,\text{mm}$, and $\theta_0 = 30^\circ$ gives $E_{\text{eff}} \approx 0.369\,\text{MPa}$. This value is an analytical prediction conditional on the provisional modulus and ideal hinge model; it is not a measured stiffness of the stent [ASSUMPTION]. The polymer material value $\nu_m \approx 0.45$ is not a substitute for the mechanism-level prediction $\nu_{\text{ideal}} = -1$.

5 Literature results and comparison with the present model

5.1 Reported results

Ali et al. measured a polyurethane auxetic film Poisson ratio between −0.87 and −0.963 over the reported uniaxial test range. Their semi-rigid stent expansion test began at a 10.00 mm diameter and 50.00 mm length; at 2.7 bar, the diameter and length reached 15.73 mm and 51.83 mm. After removal from the balloon, the stent was reported at 14.94 mm diameter and 51.8 mm length, and the stent subsequently broke at 2.86 bar [1, pp. 543–546].

In a separate rigid-stent test, the paper reports radial expansion from 0.4 to 2.2 mm and longitudinal extension from 0.1 to 1.14 mm over the reported pressure interval of 4 to 8 bar; the rigid stent cracked at 8.3 bar [1, p. 546]. The authors identify square edges and their corresponding hinges as stress-concentrated, weak regions, and note that more flexible and thicker edges/hinges improved the stent's ability to withstand stress in their tests [1, pp. 545–546].

Bukhari et al. studied a pressure-activated slider–crank deployment mechanism, rather than the same angle-only rotating-squares mechanism used in the present analytical model. For the rotating-squares stent in their comparison table, the device-expanded dimensions changed from 175 mm length and 53.17 mm diameter to 205 mm and 60.96 mm, respectively. With the balloon method, the dimensions changed to 180 mm and 60.17 mm [2, p. e203, Table III].

For their numerical model, Bukhari et al. report a maximum principal strain of 66%, compared with a stated fracture-strain limit of 400% for PMC-744 polyurethane. They also report nonuniform expansion and distortion associated with loading the stent ends at only a small number of contact points [2, p. e201].

5.2 Comparison table

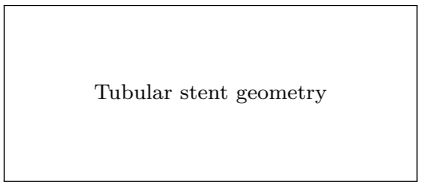
5.3 Recommended source figures and reproduction slots

The recommended choices are grouped by the purpose of the figure. Paper 1 provides a photographed tubular stent, a measured pressure–expansion plot and a visibly fractured stent. Paper 2 provides the

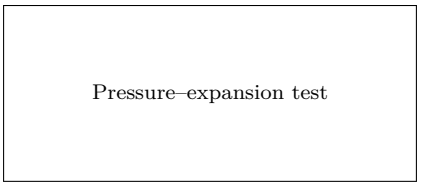
Table 3: Reported literature evidence compared with the analytical prediction for the present design. Literature values are not transferred to the present device.

Source	Expansion reported	Poisson ratio	Failure / weak regions	Relation to present model
Ali et al. [1]	Semi-rigid: 10.00 → 15.73 mm diameter at 2.7 bar; 50.00 → 51.83 mm length. Rigid test: 0.4–2.2 mm radial and 0.1–1.14 mm axial change.	Film: −0.87 to −0.963.	Semi-rigid sample broke at 2.86 bar; rigid sample cracked at 8.3 bar. Hinges/solid-square edges identified as weak regions.	Experimental polymer stents; their geometry, material and test setup differ from this design.
Bukhari et al. [2]	Rotating-square stent: device method 175 → 205 mm length and 53.17 → 60.96 mm diameter; balloon method 175 → 180 mm and 53.17 → 60.17 mm.	No single numerical Poisson ratio is tabulated for the stent in the cited comparison table [VERIFY].	Numerical maximum principal strain 66%; stated PMC-744 fracture strain 400%.	Their crank–slider actuator and PMC-744 model are not the current TPU design.
Ali thesis [3]	No numeric value inserted; thesis PDF not present in the supplied files [VERIFY].	Not extracted [VERIFY].	Not extracted [VERIFY].	Use only after the thesis PDF and its figure/table locations have been checked.
Gatt et al. [5]	Finite planar and tubular rotating-square models were compared; no deployment pressure or measured expansion for this stent is tabulated in the abstract.	Not a single transferable stent value in the abstract.	No physical failure pressure reported in the abstract.	Periodic-model moduli may overestimate finite-structure values by up to 200%; this supports caution for a finite-cell tube.
Present analytical model	15.47% nominal lattice opening from 30° to 0°; diameter ratio 1.1547.	Ideal mechanism: −1.	No failure pressure predicted; bending estimate gives 35.65% ligament surface strain for 30° → 0°.	Analytical prediction only; allowable strain and TPU constitutive data are missing.

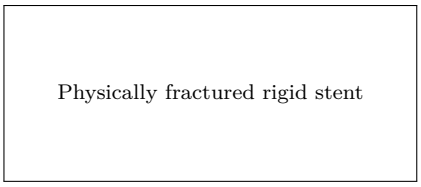
mounted stent, its measured expansion plot and a numerical strain/displacement field. The last image from Paper 2 is a numerical high-strain field, not a physical hinge-failure photograph. The thesis file could not be read; its figure numbers must therefore be found and checked before use [VERIFY].



(a) Adapted from [1], Fig. 17 (licence to be confirmed) [VERIFY]

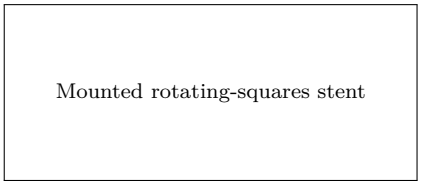


(b) Adapted from [1], Fig. 37 (licence to be confirmed) [VERIFY]

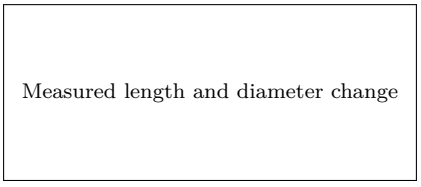


(c) Adapted from [1], Fig. 43 (licence to be confirmed) [VERIFY]

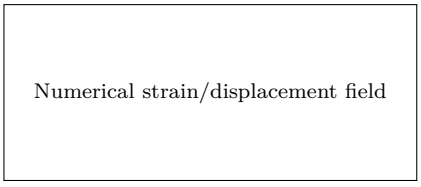
Figure 3: Suggested source figures from Paper 1 for stent geometry, expansion behaviour and physical hinge/strut failure. Each image is an adaptation slot; permission and source-page verification remain necessary.



(a) Adapted from [2], Fig. 9 (licence to be confirmed) [VERIFY]

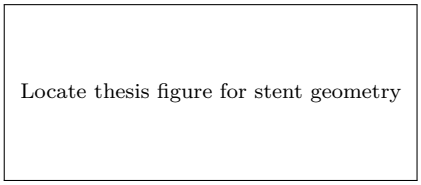


(b) Adapted from [2], Fig. 15 (licence to be confirmed) [VERIFY]

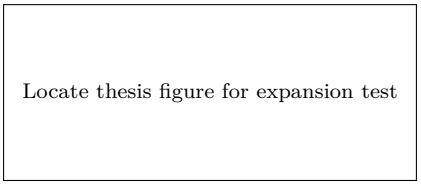


(c) Adapted from [2], Fig. 14(B–C) (licence to be confirmed) [VERIFY]

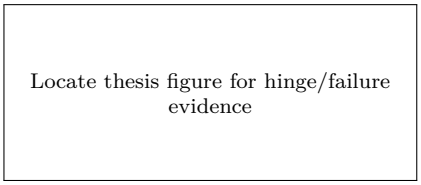
Figure 4: Suggested source figures from Paper 2. Figure 14(B–C) is a numerical field and should not be described as a physical hinge-fracture test.



(a) Adapted from [3], Fig. X (licence to be confirmed) [VERIFY]



(b) Adapted from [3], Fig. X (licence to be confirmed) [VERIFY]



(c) Adapted from [3], Fig. X (licence to be confirmed) [VERIFY]

Figure 5: Provisional figure slots for the Sheffield thesis. The thesis PDF is not readable in the supplied files, so all three figure labels remain unverified placeholders.

6 Simulation plan

6.1 Kinematic simulation using Python

The Python program in Appendix A generates an idealized unrolled pattern at 45°, 30°, 15°, and 0°; plots the diameter ratio, solid fraction and ligament surface strain against the rotation angle; and saves an animated GIF of a tubular lattice opening.

The program uses NumPy for the analytical calculations, Matplotlib for figures and animation, and Shapely for the polygonal square/ligament drawing. The unrolled-pattern drawing includes a plotting-only scale adjustment to make the hinge gaps visible; it is a kinematic illustration, not a dimensionally exact CAD reconstruction [ASSUMPTION]. The diameter, solid-fraction and ligament-strain plots use the equations in Section 4 and are analytical outputs, not solver results.

Run the program locally using Python with NumPy, Matplotlib, Shapely and Pillow installed. It writes `kinematic_pattern.png`, `diameter_ratio.png`, `solid_fraction.png`, `ligament_strain.png`, and `ring_opening.gif` to the directory containing the script. No result from these files is embedded until the user has run and checked the script.

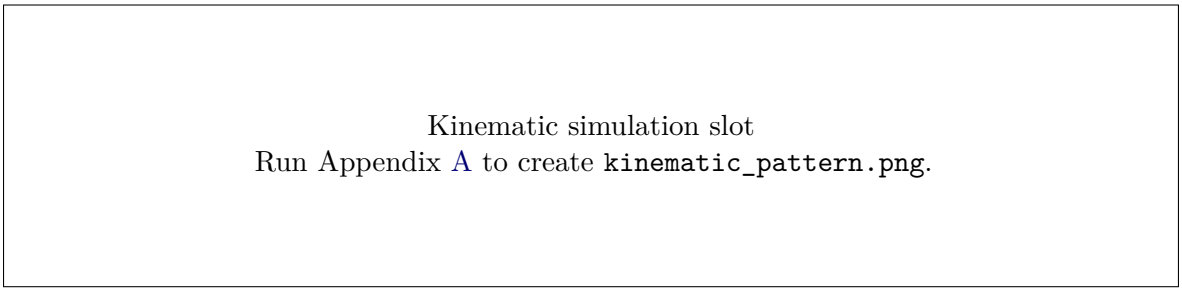
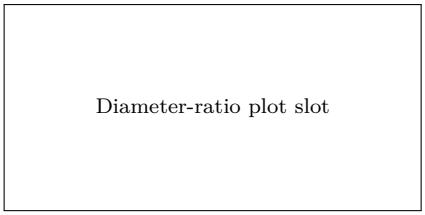
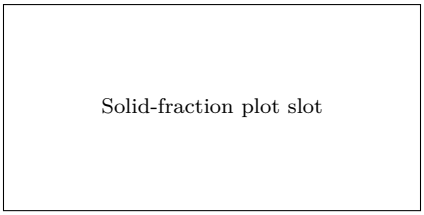


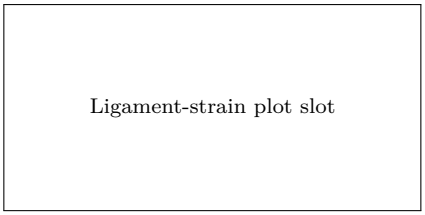
Figure 6: Unrolled rotating-squares pattern at the requested angle snapshots. Kinematic simulation to be generated by the Appendix script; idealized geometry, not an Ansys or experimental result.



(a) Diameter ratio.



(b) Solid fraction.



(c) Ligament strain.

Figure 7: Analytical plots to be generated from Eqs. (6), (4) and (7) by the Appendix script. These are analytical predictions, not finite-element results.

6.2 Finite-element analysis in Ansys

The supplied STEP geometry is a 90° sector, which is suitable for a one-sector cyclic-symmetry model if the complete stent is intended to be formed by four identical sectors [VERIFY]. The Ansys model should use the sector's cylindrical axis as the global reference and define a cylindrical coordinate system with radial, tangential and axial directions. The low- and high-angle sector faces are assigned cyclic-symmetry

constraints. The user should confirm the sector repeat count and check that the loading and requested deformation preserve cyclic symmetry.

A radial displacement is prescribed on the luminal surface in small increments. Large-deflection effects are enabled. The axial reference condition should prevent rigid-body motion without suppressing the auxetic axial extension. The radial reaction force is extracted at each imposed displacement. The effective radial-to-axial Poisson ratio is calculated from

$$\varepsilon_r = \frac{R - R_0}{R_0}, \qquad \varepsilon_z = \frac{L - L_0}{L_0}, \qquad \nu_{rz}^{\text{eff}} = -\frac{\varepsilon_z}{\varepsilon_r}. \tag{14}$$

For a first numerical check, the supplied $E \approx 26\,\text{MPa}$ and $\nu_m \approx 0.45$ can be entered as linear-elastic placeholders [ASSUMPTION]. This does not constitute a validated TPU model. A final TPU analysis should use a constitutive law fitted to test data for the actual grade and manufacturing process [VERIFY]. Mesh the ligaments with at least three elements across their width as an initial resolution target. The corresponding element width is $h/3 = 0.567/3 = 0.189\,\text{mm}$. A mesh-convergence study is still required; three elements across a narrow ligament do not by themselves prove convergence.

Table 4: Ansys result table. Leave blank until the user's own solve is complete.

Output	Value	Unit
Effective Poisson ratio, ν_{rz}^{eff}	_____	–
Radial reaction force	_____	N
Peak ligament surface strain	_____	–
Axial length change, ΔL	_____	mm

[**MY ANSYS RESULT**] Insert the final converged result set, mesh details, solver version, material model and load/displacement history here. Do not populate this box with values copied from the literature.

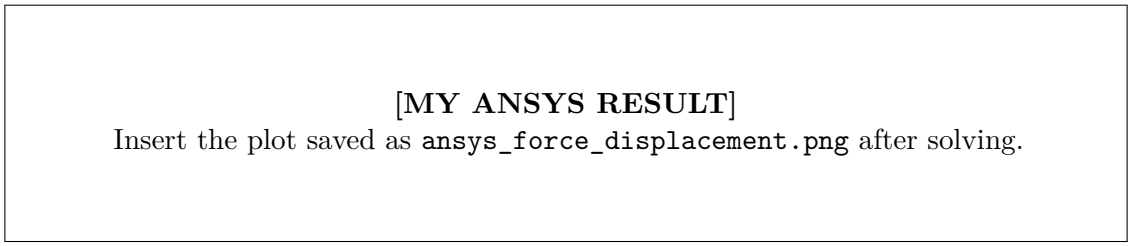


Figure 8: Ansys radial reaction force versus prescribed radial displacement. [MY ANSYS RESULT].

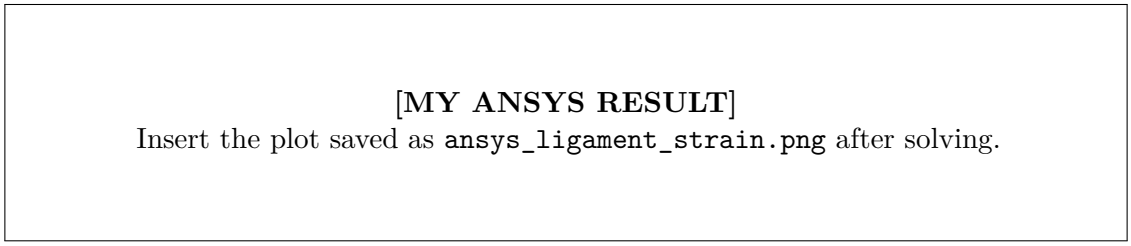


Figure 9: Ansys peak ligament strain versus prescribed radial displacement. [MY ANSYS RESULT].

6.3 Optional cross-check using FreeCAD FEM and CalculiX

A free-software cross-check can use the same imported sector and a displacement-controlled static calculation. The following recipe is intended as a starting workflow; interface labels and supported solver options depend on the installed FreeCAD and CalculiX versions [VERIFY].

1. Open the STEP sector in FreeCAD and confirm the units, radial dimensions, sector angle and axial length against Section 3. Repair invalid or open geometry before meshing.

2. Create a FEM analysis and a second-order solid mesh where practical. Refine the mesh around the ligaments and hinges. Start with an element-size target no greater than 0.189 mm across the supplied ligament width, then perform a refinement check.
3. Assign a temporary isotropic elastic material with $E = 26\,\text{MPa}$ and $\nu_m = 0.45$ only for a preliminary workflow check [ASSUMPTION]. Do not present this linear placeholder as a validated constitutive law for TPU.
4. Define the cyclic sector surfaces and use CalculiX cyclic-symmetry constraints if the installed solver build supports the required keywords. For a 90° sector, the repeat count is $N = 4$.
5. Define a cylindrical coordinate transform about the tube axis and prescribe radial displacement on the luminal node set. Constrain only the minimum additional degrees of freedom needed to prevent rigid-body motion.
6. Enable geometrical nonlinearity in the static step. Export the input deck, inspect all node/element sets and surfaces, and run the calculation. Compare displacement, reaction force, strain and mesh sensitivity with the Ansys workflow.
7. Populate the blank table below only with results generated by the user's own CalculiX run. Record the solver version, mesh element type, increment controls and any convergence warnings [MY ANSYS RESULT / user result].

A minimal input-deck template is shown below. Replace the mesh include name and all set/surface names with the identifiers actually exported by FreeCAD. The cyclic-symmetry section assumes matching sector boundary surfaces; check the generated constraints before solving [VERIFY]. The template uses a linear-elastic placeholder and is not a hyperelastic TPU material card.

Listing 1: CalculiX input-deck template; set and mesh names must be replaced.

```
** [ASSUMPTION] Units: mm, N, MPa; linear elastic placeholder only.
** [ASSUMPTION] 0.10 mm is an example radial displacement, not an analysis result.
** [ASSUMPTION] Static-step controls below are starting values; tune for convergence.
*INCLUDE, INPUT=stent_mesh_and_sets.inp

*MATERIAL, NAME=TPU_PLACEHOLDER
*ELASTIC
26., 0.45
*SOLID SECTION, ELSET=STENT_SOLIDS, MATERIAL=TPU_PLACEHOLDER

** Sector faces: replace these with exported element-face definitions.
*SURFACE, NAME=SECTOR_LOW, TYPE=ELEMENT
** <low-side element labels and face IDs>
*SURFACE, NAME=SECTOR_HIGH, TYPE=ELEMENT
** <high-side element labels and face IDs>
*TIE, NAME=CYCLIC_TIE, CYCLIC SYMMETRY
SECTOR_HIGH, SECTOR_LOW
*CYCLIC SYMMETRY MODEL, N=4, TIE=CYCLIC_TIE, NGRAPH=4
0., 0., 0., 0., 0., 1.

** Use local radial DOF 1 on the cylindrical radial-load node set.
*TRANSFORM, NSET=RADIAL_LOAD, TYPE=C
0., 0., 0., 0., 0., 1.

*STEP, NLGEOM=YES
*STATIC
0.01, 1.0, 1.e-5, 0.05
*BOUNDARY
RADIAL_LOAD, 1, 1, 0.10
** Add only the minimum rigid-body constraints needed after checking the model.
*NODE FILE
U
*EL FILE
S, E
*END STEP
```

Table 5: Optional CalculiX cross-check result table. Fill in only after running the solver.

Output	Value	Unit
Effective Poisson ratio	_____	–
Radial reaction force	_____	N
Peak ligament strain	_____	–
Axial length change	_____	mm

7 Failure and durability considerations

The rotating-squares mechanism concentrates bending in the connecting ligaments. The supplied pure-bending model predicts a ligament surface strain of approximately 35.65% when the current design rotates

from 30° to 0°. The ratio $L_h/h = 1.47$ therefore deserves particular attention. A material-grade-specific allowable strain is required before this value can be interpreted as acceptable or unacceptable [VERIFY].

Ali et al. identify the square edges and hinges as stress-concentrated weak regions. In their experiments, the semi-rigid stent broke at 2.86 bar, while the rigid stent cracked at 8.3 bar; these pressures belong to their specimens and cannot be used as failure pressures for the present design [1, pp. 545–546]. The same paper shows a numerical ring model whose failure state was inferred from stress/strain output at a high imposed radial displacement; it is not a fatigue test [1, pp. 552–553]. Bukhari et al. report a numerical peak strain and discuss nonuniform loading, but their PMC-744 material and loading mechanism are also different from this design [2, p. e201].

The material choice is provisional. TPU is specified here with approximate elastic properties, but TPU grades vary and may show nonlinear, rate-dependent and temperature-dependent behaviour [VERIFY]. A final model requires tensile or biaxial test data for the exact grade, plus a suitable constitutive fit. Fatigue, stress relaxation, creep, sterilization effects, fluid exposure, wear and cyclic deployment have not been established for the present stent [VERIFY].

8 Limitations and open questions

- The analytical model assumes ideal square units and hinge-dominated motion. It omits ligament axial stretching, local stress concentration, out-of-plane buckling, contact, friction, manufacturing imperfections and finite-strut effects [ASSUMPTION].
- The STEP file is a quarter-tube sector with a 17.00 mm axial extent. The user should confirm that four identical sectors are intended and resolve the difference from the approximate 17.3 mm target [VERIFY].
- The calculated TPU material properties are placeholders. The grade, strain-rate dependence, constitutive response and allowable hinge strain remain unverified [VERIFY].
- The allowable strain $\varepsilon_{\text{allow}}$, target deployment diameter, required radial support force, and intended clinical anatomy have not been supplied [VERIFY].
- The ideal Poisson ratio -1 is a mechanism prediction. The finite stent may show a different effective value because it contains a limited number of cells and may deform nonuniformly [VERIFY].
- The literature studies use different stent dimensions, polyurethane formulations, manufacturing methods and loading devices. Their failure pressures and strains are comparisons, not design allowables for this stent.
- The Ali thesis PDF was not present among the accessible uploads. Its exact numerical results and figure numbers must be checked against the thesis before final submission [VERIFY]. The uploaded `srep08395(1).pdf` is a different paper on hierarchical auxetic metamaterials and was not substituted for the thesis.

9 Conclusions

The supplied equations describe an ideal one-degree-of-freedom rotating-squares mechanism with equal nominal strains in both lattice directions and an ideal Poisson ratio of -1. For the supplied reference angle of 30°, the analytical model predicts a maximum nominal opening strain of 15.47% from 30° to 0°, while the pure-bending estimate gives a ligament surface strain of approximately 35.65%. The supplied ligament ratio, $L_h/h = 1.47$, therefore requires material-specific checking before the design can be considered mechanically acceptable.

The STEP file confirms the specified radial envelope and wall thickness but contains a quarter-tube sector and an axial extent of 17.00 mm, rather than a complete tube of the approximate nominal length. Literature reports establish that rotating-squares auxetic stents have been manufactured and expanded in laboratory studies, but their expansion and failure values cannot be transferred directly to the present TPU design. Ansys and optional CalculiX analyses remain to be completed; no numerical solver output is claimed here.

Key points to explain in person

What is the design? It is a tube made from square-shaped solid pieces linked by narrow flexible hinges. When the squares rotate, the gaps and overall size change.
Why is it auxetic? In the ideal mechanism, the tube's lattice expands in two directions together. Its ideal Poisson ratio is -1, meaning that the two in-plane strains have the same magnitude and sign. This is a geometry effect, not the TPU polymer's own Poisson ratio.
What does the calculation say? Starting from the supplied 30° reference angle, the ideal opening to 0° gives about 15.47% lattice strain. The hinge bending estimate is about 35.65% surface strain, so the hinges are the critical locations to examine.
What did the STEP file show? It contains one quarter of the cylindrical wall, with the requested inner and outer radii, and a length of 17.00 mm. The full-tube repetition and final length need confirmation.
What has not been proved? No Ansys or CalculiX result, physical test, fatigue life or clinical benefit is claimed. The TPU grade and allowable strain must be measured or verified before a design decision is made.

References

[1] M. N. Ali, J. J. C. Busfield, and I. U. Rehman, "Auxetic oesophageal stents: structure and mechanical properties," *Journal of Materials Science: Materials in Medicine*, vol. 25, no. 2, pp. 527–553, 2014, doi: 10.1007/s10856-013-5067-2.

[2] F. Bukhari, U. Ansari, M. N. Ali, H. Akhtar, S. Asif, U. Mohammad, and M. Mir, "A biaxial strain-based expansion mechanism for auxetic stent deployment," *Journal of Applied Biomaterials & Functional Materials*, vol. 15, no. 3, pp. e196–e205, 2017, doi: 10.5301/jabfm.5000326.

[3] M. N. Ali, "Development of Auxetic Polymeric Stent-graft for the palliative Treatment of Oesophageal Cancer," Ph.D. thesis, University of Sheffield, Sheffield, UK, 2012.

[4] J. N. Grima and K. E. Evans, "Auxetic behavior from rotating squares," *Journal of Materials Science Letters*, vol. 19, no. 17, pp. 1563–1565, 2000.

[5] R. Gatt, R. Caruana-Gauci, D. Attard, A. R. Casha, W. Wolak, K. Dudek, L. Mizzi, and J. N. Grima, "On the properties of real finite-sized planar and tubular stent-like auxetic structures," *Physica Status Solidi B*, vol. 251, no. 2, pp. 321–327, 2014, doi: 10.1002/pssb.201384257.

Completion checklist

- ☐ Add student, department, supervisor, institution and submission date to the title page.
- ☐ Confirm that the STEP quarter-sector is intended to represent one of four repeated sectors; if not, provide the full-tube STEP file. [VERIFY]
- ☐ Resolve the nominal length (approximately 17.3 mm) versus the STEP axial extent (17.00 mm). [VERIFY]
- ☐ Verify the STEP pattern against a, d, h, n_c and n_a; only the bounding envelope, wall radii, angular sector and length were checked. [VERIFY]
- ☐ Confirm the actual TPU grade and replace the provisional E and ν_m with measured or supplier-supported data. [VERIFY]
- ☐ Determine the allowable ligament strain and check the predicted 35.65% bending strain against it. [VERIFY]
- ☐ Obtain and inspect Ali's Sheffield thesis PDF; add verified figure numbers, page references and any reported numerical results. [VERIFY]
- ☐ Check the complete Bukhari article copy and its figure/page references; the uploaded .crdownload file was unreadable, so Paper 2 values were checked against the DOI full text. [VERIFY]
- ☐ Confirm permission/licensing before inserting adapted figures from the papers; replace the three source placeholders with approved images.
- ☐ Export a clean CAD view to `stent_cad_view.png`; replace the CAD placeholder.

- ☐ Run Appendix A; inspect the PNG plots and GIF, then add them using the supplied filenames.
- ☐ Complete the Ansys cyclic-symmetry model and mesh-convergence study; fill the [MY ANSYS RESULT] fields only after solving.
- ☐ Optionally run the FreeCAD–CalculiX cross-check and record its solver version, mesh and convergence information.
- ☐ Check references, figure permissions, all [VERIFY] notes and final page count before submission.

A Python script for the kinematic pattern, plots and GIF

The script below is included in the report so it can be copied into a separate file named `rotating_squares_stent_kinematics.py`. It must be run by the student. Its pattern drawing is an idealized kinematic visualization; the three curves are analytical calculations from the equations in this report. The GIF is not a finite-element or experimental result.

```python
# -*- coding: utf-8 -*-
"""Ideal kinematic illustration for the rotating-squares auxetic stent.

Run:
    python rotating_squares_stent_kinematics.py

Dependencies:
    numpy, matplotlib, shapely, pillow

Outputs (written beside this script):
    kinematic_pattern.png
    diameter_ratio.png
    solid_fraction.png
    ligament_strain.png
    ring_opening.gif

The plots are analytical predictions from the supplied equations.  The
pattern drawing and tubular GIF are kinematic illustrations, not FEA output.
"""
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation, PillowWriter
from shapely.geometry import Polygon, LineString
from shapely.ops import nearest_points

# [ASSUMPTION] User-supplied design parameters, in mm unless noted.
A_MM = 4.534
CHAMFER_MM = 0.680
HINGE_WIDTH_MM = 0.567
THETA0_DEG = 30.0
MID_RADIUS_MM = 5.0
LENGTH_MM = 17.3
N_CIRCUMFERENTIAL = 4
N_AXIAL = 2

# [MODEL eq.] L_h = sqrt(2) d cos(theta_0); angles converted to radians.
THETA0 = np.deg2rad(THETA0_DEG)
L_H_MM = np.sqrt(2.0) * CHAMFER_MM * np.cos(THETA0)

# [ASSUMPTION] Four visual snapshots were requested; they are not measured states.
SNAPSHOT_ANGLES_DEG = (45.0, 30.0, 15.0, 0.0)

# [ASSUMPTION] The visual scale factor creates visible gaps in the pattern plot.
# It is only a drawing convention and is not a revised physical square dimension.
PLOT_SQUARE_SCALE = 0.92

OUT = Path(__file__).resolve().parent


def rotated_square(cx, cy, side, angle_deg):
    """Return a Shapely square polygon rotated about its centre."""
    half = side / 2.0
    corners = np.array([
        [-half, -half], [half, -half], [half, half], [-half, half]
    ])
    angle = np.deg2rad(angle_deg)
    rot = np.array([
        [np.cos(angle), -np.sin(angle)],
        [np.sin(angle),  np.cos(angle)],
    ])
    points = corners @ rot.T + np.array([cx, cy])
    return Polygon(points)


def draw_polygon(ax, poly, facecolor, edgecolor="black", linewidth=0.9):
    x, y = poly.exterior.xy
    ax.fill(x, y, facecolor=facecolor, edgecolor=edgecolor,
            linewidth=linewidth, alpha=0.9)


def add_link(ax, poly_a, poly_b, width):
    """Draw a visible ligament between the nearest points of two squares."""
    p_a, p_b = nearest_points(poly_a, poly_b)
    line = LineString([p_a, p_b])
    if line.length > 1.0e-9:
        link = line.buffer(width / 2.0, cap_style="round")
        draw_polygon(ax, link, facecolor="0.70", edgecolor="0.25",
                     linewidth=0.45)


```

```python
def pattern_panel(ax, theta_deg):
    """Draw a schematic unrolled array using the angle-dependent pitch."""
    theta = np.deg2rad(theta_deg)
    # [MODEL eq.] D(theta)=a cos(theta); diagonal-neighbour distance=sqrt(2)D.
    d = A_MM * np.cos(theta)
    pitch = d
    side = PLOT_SQUARE_SCALE * A_MM  # [ASSUMPTION] plotting-only scale factor
    # [MODEL eq.] The drawing uses the x- and y-components of the diagonal neighbour vector D(1,1).
    # The orientation angle is chosen to show the change in square rotation.
    orientation = 45.0 + theta_deg

    polys = {}
    for j in range(N_AXIAL):
        for i in range(N_CIRCUMFERENTIAL):
            cx = (i + 0.5) * pitch
            cy = (j + 0.5) * pitch
            polys[(i, j)] = rotated_square(cx, cy, side, orientation)

    # Draw ligaments between nearest neighbours after all square faces are known.
    for j in range(N_AXIAL):
        for i in range(N_CIRCUMFERENTIAL):
            if i + 1 < N_CIRCUMFERENTIAL:
                add_link(ax, polys[(i, j)], polys[(i + 1, j)], HINGE_WIDTH_MM)
            if j + 1 < N_AXIAL:
                add_link(ax, polys[(i, j)], polys[(i, j + 1)], HINGE_WIDTH_MM)

    for poly in polys.values():
        draw_polygon(ax, poly, facecolor="#dbeafe", edgecolor="#1f2937")

    ax.set_aspect("equal", adjustable="box")
    ax.set_xlim(0.0, N_CIRCUMFERENTIAL * pitch)
    ax.set_ylim(0.0, N_AXIAL * pitch)
    ax.set_title(rf"$\theta={theta_deg:.0f}^\circ$")
    ax.set_xlabel("Unrolled circumferential coordinate (schematic)")
    ax.set_ylabel("Unrolled axial coordinate (schematic)")
    ax.grid(alpha=0.18)


def save_pattern_figure():
    fig, axes = plt.subplots(2, 2, figsize=(10, 7.2))
    for ax, theta_deg in zip(axes.flat, SNAPSHOT_ANGLES_DEG):
        pattern_panel(ax, theta_deg)
    fig.suptitle("Rotating-squares unrolled pattern (kinematic illustration)")
    fig.tight_layout()
    fig.savefig(OUT / "kinematic_pattern.png", dpi=160)
    plt.close(fig)


def save_analytical_plots():
    # [ASSUMPTION] Angle sampling is a plotting resolution, not experimental data.
    theta_deg = np.linspace(0.0, 45.0, 181)
    theta = np.deg2rad(theta_deg)

    # [MODEL eq.] Diameter ratios relative to the supplied reference and the
    # fully closed state, both derived from cos(theta)/cos(theta_reference).
    ratio_ref = np.cos(theta) / np.cos(THETA0)
    ratio_closed = np.cos(theta) / np.cos(np.deg2rad(45.0))
    fig, ax = plt.subplots(figsize=(6.4, 4.2))
    ax.plot(theta_deg, ratio_ref, label=r"$D(\theta)/D(30^\circ)$")
    ax.plot(theta_deg, ratio_closed, "--", label=r"$D(\theta)/D(45^\circ)$")
    ax.set_xlabel(r"Square angle, $\theta$ (degrees)")
    ax.set_ylabel("Diameter ratio")
    ax.set_title("Analytical diameter ratio")
    ax.grid(True, alpha=0.3)
    ax.legend()
    fig.tight_layout()
    fig.savefig(OUT / "diameter_ratio.png", dpi=160)
    plt.close(fig)

    # [MODEL eq.] Solid fraction phi_s = 1/[2 cos(theta)^2].
    phi = 1.0 / (2.0 * np.cos(theta) ** 2)
    fig, ax = plt.subplots(figsize=(6.4, 4.2))
    ax.plot(theta_deg, phi)
    ax.set_xlabel(r"Square angle, $\theta$ (degrees)")
    ax.set_ylabel("Solid fraction")
    ax.set_title("Analytical solid fraction")
    ax.grid(True, alpha=0.3)
    fig.tight_layout()
    fig.savefig(OUT / "solid_fraction.png", dpi=160)
    plt.close(fig)

    # [MODEL eq.] Ligament surface strain = |theta-theta0| h/L_h.
    eps_lig = np.abs(theta - THETA0) * HINGE_WIDTH_MM / L_H_MM
    fig, ax = plt.subplots(figsize=(6.4, 4.2))
    ax.plot(theta_deg, 100.0 * eps_lig)
    ax.axvline(THETA0_DEG, linestyle="--", linewidth=1.0,
               label=r"Reference angle $\theta_0$")
    ax.set_xlabel(r"Square angle, $\theta$ (degrees)")
    ax.set_ylabel("Ligament surface strain (%)")
    ax.set_title("Analytical ligament bending strain")
    ax.grid(True, alpha=0.3)
    ax.legend()
    fig.tight_layout()
    fig.savefig(OUT / "ligament_strain.png", dpi=160)
    plt.close(fig)


def save_ring_opening_gif():
    """Animate an idealized tubular grid whose radius follows the cosine law."""
    from mpl_toolkits.mplot3d import Axes3D  # noqa: F401; registers 3-D projection

    fig = plt.figure(figsize=(6.4, 5.4))
    ax = fig.add_subplot(111, projection="3d")
    # [ASSUMPTION] A dense angular polyline is for smooth drawing only.
    arc = np.linspace(0.0, 2.0 * np.pi, 121)
```

```python
    # [ASSUMPTION] Frame angles sweep from the closed limit to the open limit.
    frames_deg = np.linspace(45.0, 0.0, 61)
    # [ASSUMPTION] Use the nominal length as the illustrated axial span.
    z_levels = np.linspace(0.0, LENGTH_MM, N_AXIAL + 1)
    angular_nodes = np.linspace(0.0, 2.0 * np.pi,
                                N_CIRCUMFERENTIAL, endpoint=False)

    def update(theta_deg):
        ax.clear()
        theta = np.deg2rad(theta_deg)
        # [MODEL eq.] R(theta)/R(theta0)=cos(theta)/cos(theta0).
        radius = MID_RADIUS_MM * np.cos(theta) / np.cos(THETA0)

        for z in z_levels:
            ax.plot(radius * np.cos(arc), radius * np.sin(arc),
                    np.full_like(arc, z), linewidth=1.0)

        for phi in angular_nodes:
            x = radius * np.cos(phi)
            y = radius * np.sin(phi)
            ax.plot([x, x], [y, y], [0.0, LENGTH_MM], linewidth=1.0)

        # Diagonal links across each axial bay make the tube read as a lattice.
        for row in range(N_AXIAL):
            z0 = z_levels[row]
            z1 = z_levels[row + 1]
            for k, phi in enumerate(angular_nodes):
                phi_next = angular_nodes[(k + 1) % len(angular_nodes)]
                ax.plot([radius * np.cos(phi), radius * np.cos(phi_next)],
                        [radius * np.sin(phi), radius * np.sin(phi_next)],
                        [z0, z1], linewidth=0.75)

        # [ASSUMPTION] Fixed bounds keep apparent expansion comparable per frame.
        lim = 1.22 * MID_RADIUS_MM * np.cos(THETA0) / np.cos(np.deg2rad(45.0))
        ax.set_xlim(-lim, lim)
        ax.set_ylim(-lim, lim)
        ax.set_zlim(0.0, LENGTH_MM)
        ax.set_box_aspect((1.0, 1.0, 1.3))
        ax.set_xlabel("X (mm)")
        ax.set_ylabel("Y (mm)")
        ax.set_zlabel("Axial coordinate (mm)")
        ax.set_title(rf"Idealized ring opening: $\theta={theta_deg:.1f}^\circ$")
        return []

    anim = FuncAnimation(fig, update, frames=frames_deg,
                         interval=80, blit=False)  # [ASSUMPTION] display interval only
    anim.save(OUT / "ring_opening.gif", writer=PillowWriter(fps=12))
    plt.close(fig)


def main():
    save_pattern_figure()
    save_analytical_plots()
    save_ring_opening_gif()
    print("Saved kinematic pattern, analytical plots and ring-opening GIF to:")
    print(OUT)


if __name__ == "__main__":
    main()
```